

GENERAL SPECIFICATION G2.3

Project Resources

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G2.3 PROJECT RESOURCES

G2.3.1 Resources for Project

- (a) The Contractor shall employ adequate manpower working on the Project at any one time. In the event that the progress of the works is behind schedule, the Contractor shall, at the instruction of the S.O., employ additional workers in order to ensure timely completion of the works. The Contractor is deemed to have allowed for in his rates for compliance with this clause.

G2.3.2 Contractor's Organisation

- (a) The Contractor shall submit a project organisation which shall provide the following information for the S.O.'s acceptance.
- (b) A diagram showing the organisational structure for the management of the contract, with locations, names and position titles of staff and their line and staff relationship. The diagram shall include associate organisations and sub – contractors and show clearly the individuals and lines of responsibilities linking the various groups. It shall also indicate the persons designated as contacts with the S.O.. If staff are not manning the positions for the entire duration of the Contract, these shall be explained in the narrative, indicating in full detail when they are to be mobilised and demobilised.
 - (i) The names and qualifications, positions and current resumes of key executives, supervisory and engineering staff to be employed full – time on the project shall be included with the submission including principals of sub – contractors; and
 - (ii) The organisation shall include the key personnel proposed by the Contractor at the time of Tender.
- (c) The S.O. will review the Contractor's proposed organisation and shall have the right to require the Contractor to make amendments deemed necessary to meet the Contract Any changes proposed by the Contractor to the accepted Project Organisation shall be subject to the approval of the S.O..
- (d) The Contractor shall provide a competent Project Management Team to be based at the work site, from the commencement until the completion of the Work, with the necessary expertise and experience to represent the Contractor, liaise with and receive instruction from the S.O. pertaining to the Contract.
- (e) The Contractor shall engage qualified and competent professionals who have the relevant experience to carry out all the Works in accordance with the Tender Documents and Tender Drawings provided throughout the Contract period and all stages of the Project.
- (f) The Contractor shall at his own expense, form a project team led by the personnel specified in the Particular Specifications and containing all necessary personnel to deliver the Works safely, to the required quality and on time.

- (g) The personnel in the Contractor's project team shall be appropriately qualified and possess the relevant track record and professional expertise and experience for the due performance of the Works under the Contract. The qualified personnel shall be registered with the relevant professional bodies. Professional staff for electrical engineering services shall also be registered Licensed Electrical Workers for the appropriate grade.
- (h) The Contractor may with written approval of the Board, make minor adjustments in the personnel deployed and in such periods as may be appropriate to ensure the efficient performance of the Works under the Contract.
- (i) Except as the Board may otherwise agree, no change shall be made in the personnel proposed by the Contractor. If for any reason beyond the reasonable control of the Contractor, it becomes necessary to replace any of the personnel, the Contractor shall forthwith provide an appropriate replacement acceptable to the Board.
- (j) If any person specified in the Tender submitted by the Contractor is found by the Board to be incompetent in discharging his assigned responsibilities or is a security and / or safety risk or has misconducted himself, the Board may request the Contractor to forthwith provide a replacement with such qualifications and experience as may be acceptable to the Board.
- (k) Except as the Board may otherwise agree, the Contractor shall bear all costs arising out of or incidental to any replacement of personnel.
- (l) The Board undertakes no responsibility in respect of any life, health, accident, travel and other insurances, which may be necessary or desirable for the personnel or specialists associated with the Contractor for the purposes of the Project.
- (m) The Contractor shall nominate a suitably qualified and experienced engineer from his staff to be Project Manager. The nominee shall be subject to the acceptance of the S.O. who shall have the right to demand his replacement at any time after the work commences should the S.O. believe this to be in the best interest of the Project.
- (n) The Project Manager shall be continuously on the Site and devote himself full time to the Project, commencing no later than fourteen (14) calendar days from the date of the Letter of Acceptance and finishing only after the S.O. agrees in writing and has certified the Works to be fully completed. The Contractor shall also provide intermittent attendance of the Project Manager as and when required by the S.O. during the Defects Liability Period.

G2.3.3 Contractor's Qualified Person

- (a) Detailed Design by Qualified Persons
 - (i) The Contractor shall be responsible for the detailed design of all temporary works including ERSS required for the Works as well as any other structures or permanent works as indicated in the specifications to be designed and constructed by the Contractor.

- (ii) The Contractor shall engage Qualified Persons for Structural and Geotechnical (QP) to undertake the design [QP (D)] of all ERSS and temporary works. The QP engaged by the Contractor shall belong to a Professional Engineering Firm registered with the Professional Engineers Board (PEB) of Singapore. The professional engineering firm must be licensed to practise in Singapore in the relevant disciplines. A copy of the valid practicing certificate(s) shall be submitted to the S.O..
 - (iii) The appointment of the QPs by the Contractor shall be subject to the S.O.'s acceptance. Regardless of any prior acceptance, if the S.O. considers the performance of the QPs engaged by the Contractor to be unsatisfactory, the Contractor shall propose a replacement at his own expense for the acceptance of the S.O..
 - (iv) The QP shall endorse the design of all structural elements of the Works, which are designed by the Contractor (including all temporary works) as required and shall carry out all necessary submissions, via the S.O. to the appointed Accredited Checking Organisation (ACO) and the Building & Construction Authority (BCA). Where required by the Board, S.O., BCA, Building Control Acts and Regulations, the QP for Geotechnical shall also endorse all geotechnical elements of the works.
 - (v) For ERSS works, the Contractor shall submit endorsed design calculations, drawings, impact analyses on nearby structures and surrounding grounds, reports, monitoring results and interpretations, analyses, etc. for all the temporary works, open trenches / excavated slopes, temporary shafts, soil stabilization and tunnelling works including regular site visits and certifying adequacy of all ERSS works.
- (b) Supervisory Works
- (i) The Qualified Person engaged by the Contractor for the design of the Works shall be responsible for the proper supervision and inspection of all the ERSS and temporary works necessary to construct the permanent works designed by the S.O..
 - (ii) The Contractor shall deploy an adequate number of qualified and experienced supervisory staff (Qualified Site Supervisors per the requirements of the Building Control Act and Regulations) to ensure that the Works are carried out in accordance with the approved drawings and specifications and to the satisfaction of the S.O.. The Contractor's appointed QP shall be responsible for endorsing all necessary forms to obtain permit to commence work from BCA.
 - (iii) The Contractor's QP shall maintain his professional independence from the Contractor's site staff, site managers etc. and shall report directly to the Contractor's Project Director.

G2.3.4 Employment of Certified Skilled Workers

- (a) The Contractor shall employ certified skilled workers for the various trades required for the Works. These certified workers shall possess one or more of the certificates for relevant trades issued by the Building and Construction Authority (BCA), Institute of Technical Education (ITE) or other approved local institutions.

- (b) The Contractor shall produce original copies of the relevant trade certificates of the workers as and when requested by the S.O..
- (c) The Contractor shall be deemed to have included in his rates for compliance with the requirements stipulated in this clause.

END OF SECTION